

# Al-rich AlGaN based transistors

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## ABSTRACT

Research results for AlGaN-channel transistors are reviewed as they have progressed from low Al-content and long-channel devices to Al-rich and short-channel RF devices. Figure of merit (FOM) analysis shows encouraging comparisons relative to today's state-of-the-art GaN devices for high Al-content and elevated temperatures. Critical electric field ( $E_C$ ), which fuels the AlGaN transistor FOM for high Al-composition, is not measured directly, but average gate-drain electric field at breakdown is substantially better in multiple reported AlGaN-channel devices compared to GaN. Challenges for AlGaN include the constraints arising from relatively low room temperature mobility dominated by ternary alloy scattering and the difficulty of making low-resistivity Ohmic contacts to high Al-content materials. Nevertheless, considerable progress has been made recently in the formation of low-resistivity Ohmic contacts to Al-rich AlGaN by using reverse compositional grading in the semiconductor, whereby a contact to a lower-Al alloy (or even to GaN) is made. Specific contact resistivity ( $\rho_c$ ) approaching  $\rho_c \sim 2 \times 10^{-6} \Omega \text{ cm}^2$  to AlGaN devices with 70% Al-content in the channel has been reported. Along with scaling of the channel length and tailoring of the threshold voltage, this has enabled a dramatic increase in the current density, which has now reached 0.6 A/mm. Excellent  $I_{ON}/I_{OFF}$  current ratios have been reported for Schottky-gated structures, in some cases exceeding  $10^9$ . Encouraging RF performance in Al-rich transistors has been reported as well, with  $f_T$  and  $f_{max}$  demonstrated in the tens of gigahertz range for devices with less than 150 nm gates. Al-rich transistors have also shown lesser current degradation over temperature than GaN in extreme high-temperature environments up to 500 °C, while maintaining  $I_{ON}/I_{OFF}$  ratios of  $\sim 10^6$  at 500 °C. Finally, enhancement-mode devices along with initial reliability and radiation results have been reported for Al-rich AlGaN transistors. The Al-rich transistors promise to be a very broad and exciting field with much more progress expected in the coming years as this technology matures.

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## I. INTRODUCTION

The earliest transistors<sup>1</sup> used Ge with a bandgap of 0.66 eV. The difficulty of purifying Ge and its temperature sensitive electronic properties led to the search for other semiconductors. The Si bipolar junction transistor<sup>2,3</sup> with a bandgap of 1.1 eV proved to be superior to Ge on both counts, and a modern version of it known as the insulated gate bipolar transistor is still a mainstay in a segment of today's power electronics market. Elucidation of the methods and details of Si oxidation and doping<sup>4</sup> led to the development of the metal-oxide-semiconductor field effect transistor (MOSFET).<sup>5</sup> Its many variants are ubiquitous in modern electronics, including power electronics at voltages below 600 V.<sup>6,7</sup> High electron mobility transistors (HEMTs)<sup>8</sup> and heterojunction bipolar transistors (HBTs)<sup>9</sup> were first developed on GaAs with a bandgap of 1.42 eV. They found niche uses in high frequency radio-frequency (RF) electronics but not in power electronics applications associated with high breakdown voltage, high electric field operation. GaP/AlGaP HBTs

with a base bandgap near 2.0 eV are an example of a candidate for high-temperature electronics<sup>10</sup> that never gained traction beyond the research stage. SiC with a bandgap of 3.25 eV and GaN with a bandgap of 3.4 eV became the next frontier of research to capitalize on the properties of wide bandgap (WBG) semiconductors. The critical electric field ( $E_C$ ) of semiconductors scales exponentially with the bandgap ( $E_G$ ) as approximately  $E_C \sim E_G^{2.5}$  (Ref. 11) and is approximately 10× larger for SiC and GaN compared to Si and GaAs. These WBG semiconductors ushered in the next era of high voltage, high power device research that was able to demonstrate compelling performance early in the development cycle. Market entry in both RF and power electronics was more gradual as cost and reliability metrics were in competition with a continually improving Si-based technology. Today, the WBG-based electronics have found large-scale market adoption.

The first SiC MOSFETs were demonstrated in the late 1980s,<sup>7,12</sup> followed by power MOSFETs in 1994.<sup>13</sup> Blocking voltages increased 75% per year in the late 1990s reaching 6.1 kV by 2000.<sup>14,15</sup> At this